

Part II — Motion and Forces · Chapter 9

Projectile Motion

Analyzing two-dimensional motion under gravity by treating horizontal and vertical components as two independent one-dimensional problems

A ball thrown horizontally from a table and a ball simply dropped from the same height land at exactly the same time. A bullet fired from a horizontal rifle hits the ground at the same instant as a bullet dropped from the same height. Why? Because horizontal and vertical motions are completely independent. This single insight converts every projectile problem — no matter how complex it looks — into two linked problems from Chapter 8.

9.1 What Makes Motion a Projectile

A projectile is any object that is launched into the air and then moves under the influence of gravity alone, with no propulsion. Once the object leaves the launching mechanism — whether a hand, a cannon, or a ramp — it becomes a projectile. The word ‘projectile’ does not imply a weapon; it applies to a thrown baseball, a water droplet from a fountain, a long-jumper in the air, or a package dropped from an airplane.

The defining feature of ideal projectile motion is that the only force acting on the object after launch is gravity. This means the vertical acceleration is a constant -9.81 m/s^2 (with upward positive) and the horizontal acceleration is exactly zero. These two facts, taken together, completely determine the trajectory.

Key Point: Real-world projectiles also experience air resistance, spin, and the Coriolis effect from Earth’s rotation. We ignore all of these in this chapter — Section 9.13 discusses when those corrections matter and how large they are.

9.2 Assumptions of Ideal Projectile Motion

The analysis in this chapter rests on three idealizations. State them explicitly at the start of every projectile problem.

- No air resistance: the only force after launch is gravity. This is accurate for dense, compact objects at moderate speeds (a shot put, a baseball, a golf ball at short range) and breaks down for light, wide, or fast objects.

- Constant gravitational acceleration: $g = 9.81 \text{ m/s}^2$ downward, uniform everywhere during the flight. This is valid as long as the height change is much smaller than Earth's radius ($\approx 6400 \text{ km}$). For any projectile problem in this book, the assumption holds easily.
- Flat Earth approximation: the curvature of Earth's surface is negligible over the range of the projectile. Valid for ranges much less than $\approx 8 \text{ km}$.

Historical Note: In 1638, Galileo derived the parabolic trajectory of a projectile by separating motion into independent horizontal and vertical components — precisely the method we use today. His *Two New Sciences* established both kinematics and the independence principle decades before Newton published his laws.

9.3 Horizontal and Vertical Motion Are Independent

The foundational principle of projectile analysis: the horizontal and vertical components of motion are completely independent. Each obeys its own set of kinematic equations. They share only one thing: time. The time elapsed is the same for both components.

This independence follows directly from Newton's second law applied separately to each direction. Gravity acts only downward; it has zero horizontal component. No force acts horizontally (by our assumptions); so there is zero horizontal acceleration. The two directions do not communicate with each other.

The Rule: To solve any projectile problem: (1) decompose the launch velocity into x and y components; (2) solve the horizontal problem as constant velocity; (3) solve the vertical problem as constant acceleration (free fall); (4) use time as the link between the two.

HORIZONTAL (x-direction)	VERTICAL (y-direction)
$a_x = 0$ (no force)	$a_y = -9.81 \text{ m/s}^2$ (gravity)
$v_x = v_i \cos\theta = \text{constant}$	$v_y = v_i \sin\theta + a_y t$
$x = v_x \cdot t$	$y = v_{y0} t + \frac{1}{2} a_y t^2$

Figure 9.1 The two independent component equations. Time t is the same in both.

9.4 Launch Velocity and Vector Components

The launch velocity is a vector with magnitude v_i (the launch speed) and direction θ above the horizontal. As in Chapter 4, we decompose it into horizontal and vertical components:

$$v_{x0} = v_i \cdot \cos\theta \quad v_{y0} = v_i \cdot \sin\theta$$

v_{x0} is the initial horizontal velocity; v_{y0} is the initial vertical velocity. θ is measured from the horizontal.

These components are the starting values for the two independent kinematic analyses. The horizontal component v_{x0} stays constant throughout the flight. The vertical component v_{y0} changes every instant due to gravitational acceleration.

Launch Angle θ	$v_{x0} = v_i \cos\theta$	$v_{y0} = v_i \sin\theta$	Character of Motion
0° (horizontal)	$= v_i$ (full)	$= 0$	Horizontal launch; drops immediately
30°	$= 0.866 v_i$	$= 0.500 v_i$	More horizontal than vertical
45°	$= 0.707 v_i$	$= 0.707 v_i$	Equal horizontal and vertical; max range (no air resistance)
60°	$= 0.500 v_i$	$= 0.866 v_i$	More vertical than horizontal; higher, shorter range
90° (vertical)	$= 0$	$= v_i$ (full)	Straight up; zero horizontal range

Table 9.1 Component breakdown for common launch angles. Note complementary angles (30°/60°) give the same range.

Common Mistake: The angle θ is always measured from the horizontal, not from the vertical. If a problem gives an angle from the vertical, subtract it from 90° first.

9.5 Horizontal Motion in Projectile Problems

Because horizontal acceleration is zero, horizontal motion is uniform: constant velocity throughout the flight. The only equation needed for the horizontal direction is:

$$x = v_{x0} \cdot t \quad \text{where} \quad v_{x0} = v_i \cos\theta$$

The horizontal velocity v_{x0} never changes — it is the same at launch, at the peak, and just before landing. There is no horizontal force to speed it up or slow it down (by our assumptions).

Example — Horizontal motion only A ball is launched at 20.0 m/s horizontally (so $\theta = 0$, $v_{x0} = 20.0$ m/s). After 3.0 s: $x = 20.0 \times 3.0 = 60.0$ m. The ball has moved 60.0 m horizontally regardless of what it does vertically.

Key Point: The horizontal component of velocity is unchanged even at the very peak

of the trajectory. At the peak, $v_y = 0$, but $v_x = v_{x0}$ throughout. The speed at the peak equals v_{x0} (not zero).

9.6 Vertical Motion in Projectile Problems

Vertical motion is free fall: constant downward acceleration $g = 9.81 \text{ m/s}^2$. Using upward as positive, $a_y = -9.81 \text{ m/s}^2$. The kinematic equations from Chapter 8 apply directly:

$$\begin{aligned}v_y &= v_{y0} + a_y t \\y &= v_{y0} t + \frac{1}{2} a_y t^2 \\v_y^2 &= v_{y0}^2 + 2 a_y \Delta y\end{aligned}$$

The vertical and horizontal analyses share time t . Finding t from one component and substituting into the other is the most common technique for linking the two equations.

Example — Vertical motion Same ball launched at 20.0 m/s at 30.0° above horizontal. $v_{y0} = 20.0 \sin 30^\circ = +10.0 \text{ m/s}$. After $t = 3.0 \text{ s}$: $v_y = 10.0 + (-9.81)(3.0) = 10.0 - 29.4 = -19.4 \text{ m/s}$ (downward). $y = 10.0(3.0) + \frac{1}{2}(-9.81)(3.0)^2 = 30.0 - 44.1 = -14.1 \text{ m}$. The ball is 14.1 m below the launch point at $t = 3.0 \text{ s}$.

9.7 Time of Flight

The time of flight is the total time the projectile is in the air from launch to landing. For a projectile launched from and returning to the same height, time of flight is found by setting $y = 0$ in the vertical kinematic equation and solving for t .

$$0 = v_{y0} t + \frac{1}{2} a_y t^2 = t(v_{y0} + \frac{1}{2} a_y t)$$

This equation has two solutions: $t = 0$ (the launch instant) and $t = -2 v_{y0} / a_y = 2 v_{y0} / g$ (the landing instant). The time of flight is:

$$T = 2 v_{y0} / g = 2 v_i \sin\theta / g$$

Valid when the landing height equals the launch height.

Example — Time of flight A ball is launched at 25.0 m/s at 40.0° . $v_{y0} = 25.0 \sin 40^\circ = 16.07 \text{ m/s}$. $T = 2(16.07) / 9.81 = 3.28 \text{ s}$.

Common Mistake: This formula for T applies only when landing height = launch height. For elevated launches or drops, set up the full equation $y = v_{y0} t + \frac{1}{2} a_y t^2$ with the correct final y value and solve for t using the quadratic formula.

9.8 Maximum Height

Maximum height is reached when the vertical velocity is zero ($v_y = 0$). At that instant, the projectile is moving purely horizontally. Using the vertical kinematic equation without time:

$$0 = v_{y0}^2 + 2 a_y H \rightarrow H = v_{y0}^2 / (2g) = (v_i \sin\theta)^2 / (2g)$$

H = maximum height above the launch point. Depends only on the vertical component of launch velocity.

Example — Maximum height $v_i = 25.0$ m/s, $\theta = 40.0^\circ$. $v_{y0} = 16.07$ m/s. $H = (16.07)^2 / (2 \times 9.81) = 258.2 / 19.62 = 13.2$ m.

The time to reach maximum height is exactly half the total time of flight (for symmetric trajectories): $t_{\text{peak}} = v_{y0} / g$.

9.9 Horizontal Range

The horizontal range R is the total horizontal distance from launch to landing. It is found by substituting the time of flight into the horizontal equation:

$$R = v_{x0} \cdot T = v_i \cos\theta \cdot (2 v_i \sin\theta / g)$$

$$R = v_i^2 \sin(2\theta) / g$$

Range formula for equal launch and landing heights. Maximum range at $\theta = 45^\circ$ (where $\sin 90^\circ = 1$).

Example — Range $v_i = 25.0$ m/s, $\theta = 40.0^\circ$. $R = (25.0)^2 \sin(80^\circ) / 9.81 = 625 \times 0.9848 / 9.81 = 62.8$ m.

Key Point: Complementary angles (e.g., 30° and 60°) give the same range, because $\sin(2 \times 30^\circ) = \sin(60^\circ) = \sin(120^\circ) = \sin(2 \times 60^\circ)$. This symmetry is a direct consequence of the range formula.

Bridge Note The range formula $R = v_i^2 \sin(2\theta)/g$ is derived in Volume II of the advanced series from the Lagrangian of a particle in a gravitational field. There, it emerges naturally from variational principles rather than from decomposing equations.

9.10 Launches from Elevated Positions

When a projectile is launched from a height h above the landing point (or below it), the simple formulas for T , H , and R no longer apply directly. Instead, set up the vertical equation with the correct displacement and solve the quadratic.

Worked Example 9.1 — Horizontal Launch from a Cliff

A ball is launched horizontally ($v_i = 15.0 \text{ m/s}$, $\theta = 0^\circ$) from a cliff 45.0 m above the ground. Find: (a) time of flight; (b) horizontal range; (c) velocity at impact.

Step 1: Setup: $v_{x0} = 15.0 \text{ m/s}$; $v_{y0} = 0$ (horizontal launch); $a_y = -9.81 \text{ m/s}^2$; $\Delta y = -45.0 \text{ m}$ (falls 45 m).

Step 2: Time (vertical): $y = v_{y0} t + \frac{1}{2} a_y t^2 \rightarrow -45.0 = 0 + \frac{1}{2}(-9.81)t^2 \rightarrow t^2 = 9.174 \rightarrow t = 3.03 \text{ s}$

Step 3: Range (horizontal): $x = 15.0 \times 3.03 = 45.4 \text{ m}$

Step 4: Velocity at impact: $v_x = 15.0 \text{ m/s}$; $v_y = 0 + (-9.81)(3.03) = -29.7 \text{ m/s}$.
Speed = $\sqrt{(15.0^2 + 29.7^2)} = \sqrt{(225 + 882)} = \sqrt{1107} = 33.3 \text{ m/s}$

Step 5: Direction: $\tan \alpha = |v_y|/v_x = 29.7/15.0 = 1.98 \rightarrow \alpha = 63.3^\circ$ below horizontal

Worked Example 9.2 — Angled Launch from a Height

A ball is launched at 20.0 m/s at 30.0° above horizontal from a platform 10.0 m above the ground. Find the horizontal range.

Step 1: Components: $v_{x0} = 20.0 \cos 30^\circ = 17.32 \text{ m/s}$; $v_{y0} = 20.0 \sin 30^\circ = 10.0 \text{ m/s}$

Step 2: Vertical equation: Final $y = -10.0 \text{ m}$ (lands 10 m below launch). $-10.0 = 10.0 t + \frac{1}{2}(-9.81)t^2$

Step 3: Rearrange: $4.905 t^2 - 10.0 t - 10.0 = 0 \rightarrow t = [10.0 \pm \sqrt{(100 + 4 \times 4.905 \times 10.0)}] / (2 \times 4.905)$

Step 4: Solve: $t = [10.0 \pm \sqrt{296.2}] / 9.81 = [10.0 \pm 17.21] / 9.81 \rightarrow t = 2.776 \text{ s}$ (positive root)

Step 5: Range: $x = 17.32 \times 2.776 = 48.1 \text{ m}$

9.11 Symmetry in Projectile Motion

For a projectile launched and landing at the same height, the trajectory is perfectly symmetric about the peak. This symmetry produces several useful relationships:

- The time to reach the peak equals the time to fall from the peak back to launch height: $t_{\text{up}} = t_{\text{down}} = T/2$.
- The speed at any point on the way up equals the speed at the corresponding point on the way down at the same height.
- The launch angle below the horizontal at impact equals the launch angle above the horizontal at launch (in magnitude).
- At the peak, $v_y = 0$ and $v_x = v_{x0}$; the speed is at its minimum value of $v_i \cos \theta$.

- The vertical velocity at impact has the same magnitude as v_{y0} but opposite sign: $v_{yf} = -v_{y0}$.

Tip: Use symmetry as a check. If your calculated impact speed differs from the launch speed when landing at the same height, something is wrong — the magnitudes must match (though the direction changes). Similarly, landing time should be exactly $2 \times$ (peak time).

9.12 Projectile Motion Graphs and Diagrams

Four graphical representations capture different aspects of projectile motion.

Graph / Diagram	What It Shows	Key Features
Trajectory (y vs x)	The actual curved path in space	Parabola; peak at $T/2$; symmetric for equal heights
x-t graph (horizontal position)	Horizontal position vs time	Straight line; slope = v_{x0} (constant)
y-t graph (vertical position)	Vertical position vs time	Downward parabola; peak at $t = v_{y0}/g$
v_y vs t graph	Vertical velocity vs time	Straight line with slope -9.81 m/s^2 ; crosses zero at peak
v_x vs t graph	Horizontal velocity vs time	Horizontal line at v_{x0} (constant)

Table 9.2 Graphical representations of projectile motion and their key features.

The trajectory is a parabola. This can be shown by eliminating t between $x = v_{x0} t$ and $y = v_{y0} t + \frac{1}{2} a_y t^2$. Substituting $t = x/v_{x0}$ into the y equation gives $y = (v_{y0}/v_{x0})x + \frac{1}{2} a_y (x/v_{x0})^2$, which is a quadratic in x : the equation of a parabola.

9.13 Real-World Deviations: Air Resistance and Drag

The ideal projectile model predicts that: (1) range is maximized at 45° ; (2) complementary angles give identical ranges; (3) the trajectory is a perfect parabola. Real projectiles deviate from all three predictions whenever air resistance is significant.

Air resistance (drag) acts opposite to the velocity vector and depends on speed — typically as $F_{\text{drag}} \propto v^2$ for fast objects. This force is not constant, so the constant-acceleration equations do not apply. The effects are:

- Maximum range angle drops below 45° (typically $30\text{--}40^\circ$ for balls, less for bullets).
- Complementary angles no longer give equal ranges; shallow angles outperform steep ones.
- The trajectory is no longer symmetric: the descent is steeper than the ascent.
- Impact speed is less than launch speed (energy lost to drag).

For dense, compact objects at moderate speeds (shot put, javelin at short distances, golf ball at ranges <100 m), air resistance produces corrections of order 10–20% and the ideal model is a reasonable approximation. For lightweight objects (badminton shuttlecock, table tennis ball), air resistance dominates and the ideal model fails badly.

Projectile	Effect of Air Resistance	Ideal Model Quality
Steel shot put	Small (13%)	Excellent
Baseball (50 m)	Moderate (115%)	Good
Golf ball (full drive)	Significant (140%)	Rough estimate only
Table tennis ball	Very large (dominant)	Poor; spin effects also matter
Badminton shuttle	Extreme; decelerates rapidly	Fails completely

Table 9.3 *Air resistance effects on common projectiles. Denser, smaller, slower projectiles are better modeled as ideal.*

9.14 Common Mistakes in Projectile Problems

Mistake 1: Mixing x and y components

Never use a horizontal velocity in a vertical equation or vice versa. Keep a strict two-column layout: horizontal equations on the left, vertical on the right. Clearly label every variable with a subscript x or y.

Mistake 2: Assuming the speed at the peak is zero

Only the vertical component of velocity is zero at the peak. The horizontal component v_{x0} persists throughout. The speed at the peak equals $v_{x0} = v_i \cos\theta$ — which is non-zero for any $\theta < 90^\circ$.

Mistake 3: Forgetting to decompose the launch velocity

Substituting the full launch speed v_i directly into horizontal or vertical equations — without using $\cos\theta$ or $\sin\theta$ — is one of the most common errors. Always decompose

first.

Mistake 4: Using the range formula for elevated launches

$R = v_i^2 \sin(2\theta)/g$ applies only when the landing height equals the launch height. For any other situation, use the general vertical equation and solve for t with the quadratic formula.

Mistake 5: Getting the sign of the vertical displacement wrong

Choose upward positive and apply it consistently. If a projectile is launched from a cliff, the landing point has $y < 0$ relative to the launch point. Substituting $y = -h$ (negative) into the vertical equation gives the correct downward displacement.

Common Mistake: Draw a coordinate system with labeled positive directions before writing any equations. Mark the launch point as the origin. Write the y -coordinate of the landing point as positive or negative based on whether it is above or below the launch point.

Chapter Summary

A projectile moves in two dimensions under gravity alone. The key to every projectile problem is separating motion into independent horizontal (constant velocity) and vertical (constant acceleration) components, linked only by the shared time variable.

- Horizontal: $a_x = 0$; $v_x = v_i \cos\theta = \text{constant}$; $x = v_x t$.
- Vertical: $a_y = -9.81 \text{ m/s}^2$; $v_y = v_{y0} + a_y t$; $y = v_{y0} t + \frac{1}{2}a_y t^2$.
- At the peak: $v_y = 0$; $v_x = v_{x0}$ (unchanged); speed = $v_i \cos\theta$.
- Time of flight (same height): $T = 2v_{y0}/g = 2v_i \sin\theta/g$.
- Maximum height: $H = v_{y0}^2/(2g)$.
- Range (same height): $R = v_i^2 \sin(2\theta)/g$; maximized at $\theta = 45^\circ$.
- For elevated launches, use the quadratic formula on the full vertical equation.
- The trajectory is a parabola; symmetric about the peak for equal-height launches.
- Air resistance reduces range, lowers the optimal angle below 45° , and breaks symmetry.

Key Terms

projectile: Any object launched into the air and subsequently moving under gravity alone.

trajectory: The curved path followed by a projectile through space; a parabola under ideal conditions.

independence principle: The horizontal and vertical components of projectile motion are independent; each obeys its own kinematic equations.

v_{x0} (initial horizontal velocity): The constant horizontal component of velocity throughout the flight: $v_i \cos\theta$.

v_{y0} (initial vertical velocity): The initial vertical component of velocity at launch: $v_i \sin\theta$; changes during flight due to gravity.

time of flight: The total time a projectile is in the air from launch to landing.

maximum height: The greatest height above the launch point reached during the trajectory; occurs when $v_y = 0$.

range: The horizontal distance from the launch point to the landing point.

launch angle: The angle above the horizontal at which a projectile is launched; determines the mix of horizontal and vertical velocity.

air resistance: A drag force opposing motion through air; causes real trajectories to deviate from the ideal parabolic path.

Concept Review Questions

1. Explain in your own words why the horizontal and vertical motions of a projectile are independent.
2. A ball is launched horizontally from a cliff. Another ball is dropped from the same height at the same instant. Which lands first? Why?
3. At the peak of its trajectory, what is the vertical velocity of a projectile? What is the horizontal velocity? What is the total speed?
4. Why does a 45° launch angle maximize range when there is no air resistance?
5. Two projectiles are launched at 30° and 60° with the same speed. Compare their ranges, maximum heights, and times of flight.
6. A ball is launched horizontally from a table. Sketch the x - t , y - t , v_x - t , and v_y - t graphs for its motion from launch to landing.

7. Why does air resistance lower the optimal launch angle below 45° ?
 8. Why is the range formula $R = v_i^2 \sin(2\theta)/g$ invalid for elevated launches?
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Vector Decomposition Practice

Decompose each launch velocity into horizontal and vertical components. Round to 3 significant figures.

9. $v_i = 30.0$ m/s at 25.0° above horizontal.
 10. $v_i = 45.0$ m/s at 60.0° above horizontal.
 11. $v_i = 12.0$ m/s horizontally (0°).
 12. $v_i = 18.5$ m/s at 37.0° above horizontal.
 13. A ball leaves a bat at 35.0 m/s at 28.0° . Find both components.
 14. Given $v_x = 14.0$ m/s and $v_y = 10.0$ m/s at launch, find v_i and θ .
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Projectile Motion Problems

Assume launch and landing at the same height unless stated otherwise. Use $g = 9.81$ m/s². Show component analysis.

15. A ball is launched at 20.0 m/s at 35.0° . Find: (a) time of flight; (b) maximum height; (c) horizontal range.
 16. A soccer ball is kicked at 15.0 m/s at 40.0° . Find: (a) time to reach peak; (b) peak height; (c) range; (d) speed at peak.
 17. A long jumper leaves the ground at 8.90 m/s at 23.0° . How far does she jump?
 18. A ball is thrown at 25.0 m/s. What two launch angles give a range of 50.0 m?
 19. At what launch angle is the range equal to the maximum height?
 20. A projectile is in the air for 4.20 s and lands 120 m from the launch point. Find v_{x0} and v_{y0} at launch.
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Elevated Launch Problems

21. A ball is thrown horizontally at 12.0 m/s from a 20.0 m cliff. Find: (a) time of flight; (b) range; (c) impact speed and angle.
22. A package is dropped from an airplane traveling at 90.0 m/s horizontally at 500 m altitude. Find where it lands relative to the drop point.
23. A ball is launched at 18.0 m/s at 30.0° above horizontal from a 15.0 m platform. Find: (a) time of flight; (b) horizontal range; (c) speed at impact.
24. A stone is thrown horizontally from a bridge 40.0 m above a river. It lands 55.0 m downstream. Find: (a) the launch speed; (b) the impact angle.
25. A ball is launched at 22.0 m/s at 45.0° from the edge of a 30.0 m building. Find the total time of flight and where it lands relative to the base of the building.

Conceptual Projectile Analysis

26. Sketch the trajectory of a ball launched at 45° and at 30° with the same speed. Label which goes higher and which goes farther.
27. Explain why a bullet fired horizontally and one dropped simultaneously from the same height both hit the ground at the same time.
28. A ball is thrown upward at 20.0 m/s from a moving car traveling at 15.0 m/s. Describe the trajectory as seen by: (a) the driver; (b) a person standing on the road.
29. For a given launch speed, the maximum range angle is 45° . If a quarterback must throw over a 3.0 m defensive lineman 5.0 m away, is 45° still the best angle? Explain qualitatively.

Chapter 9 Checkpoint

I can...	Section
State the three assumptions of ideal projectile motion	9.2

Decompose a launch velocity into horizontal and vertical components	9.4
Write the correct kinematic equations for each component	9.3, 9.5, 9.6
Find the time of flight for equal-height launches	9.7
Find maximum height using the vertical kinematic equations	9.8
Find range using $R = v_i^2 \sin(2\theta)/g$ (same height only)	9.9
Solve elevated launch problems with the quadratic formula	9.10
Use trajectory symmetry to check answers	9.11
Identify how air resistance modifies ideal predictions	9.13
Avoid the five most common projectile mistakes	9.14

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